Chapter 1

Quantum Theory

Part I

Contents

- Overview
- Introduction to basic principles

Origins of Quantum Mechanics

17th Century, Isaac Newton ⇒ **Classical Mechanics**
→ Describe behaviors of **everyday-sized** objects

In the end of 19th century:
Failures of classical physics to very small particles



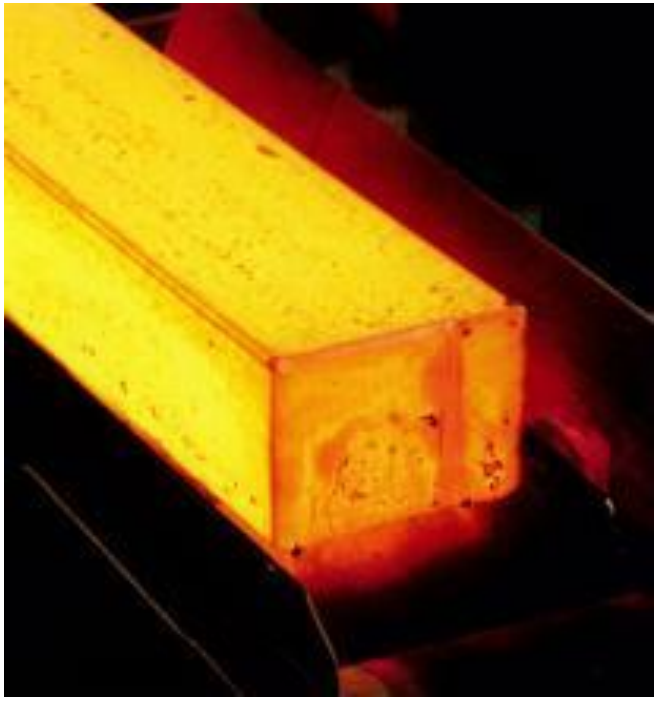
Isn't Classical mechanics quite complete?

→ **Quantum mechanics**

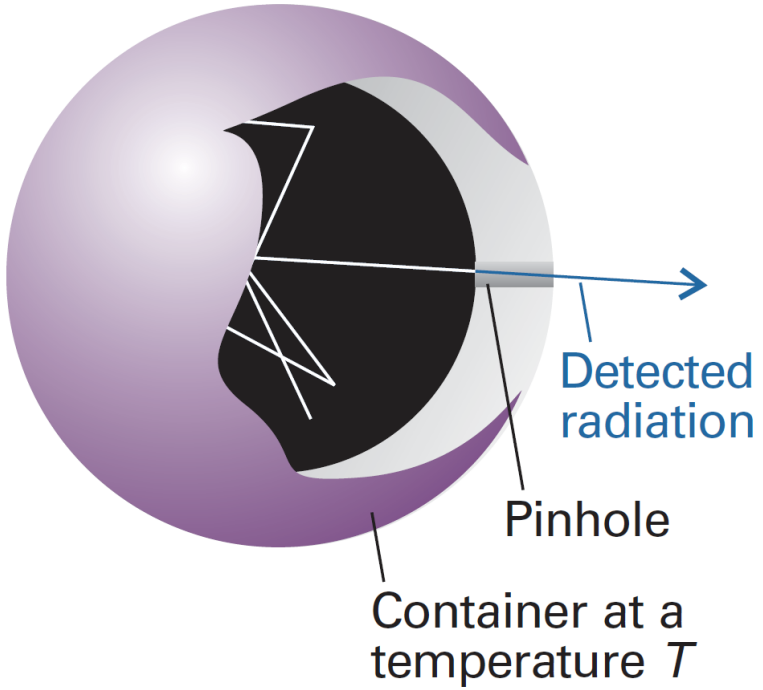
At small scales

Failures of classical physics

Black-Body Radiation: When some object is heated up, it emits radiation of varying frequencies and intensities based on its temperature.



A heated iron bar glowing red hot becomes white hot when heated further



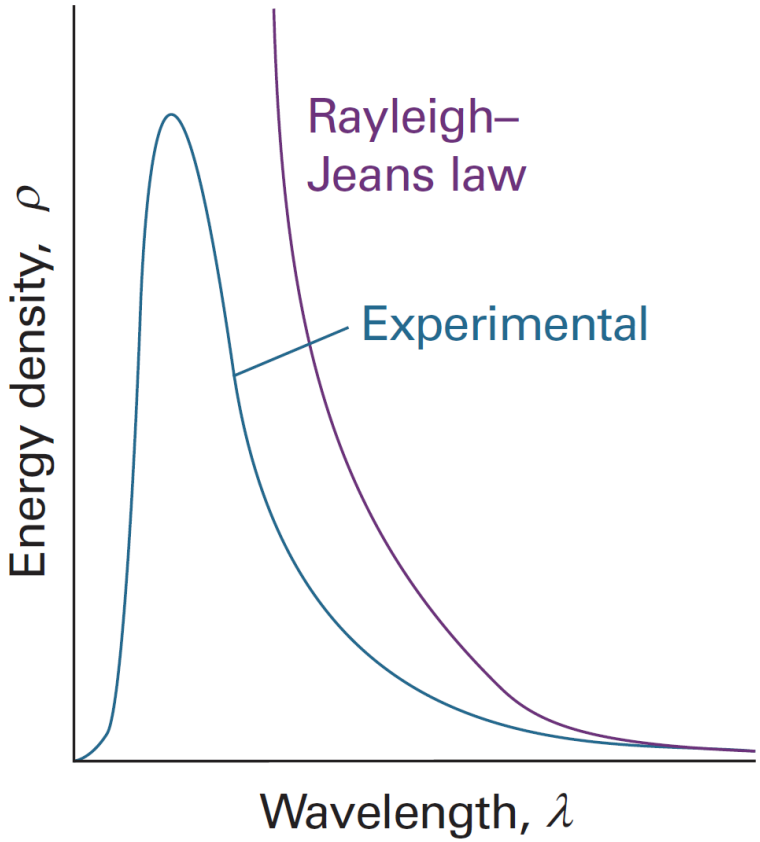
Blackbody

Classical → Raleigh-Jeans law:

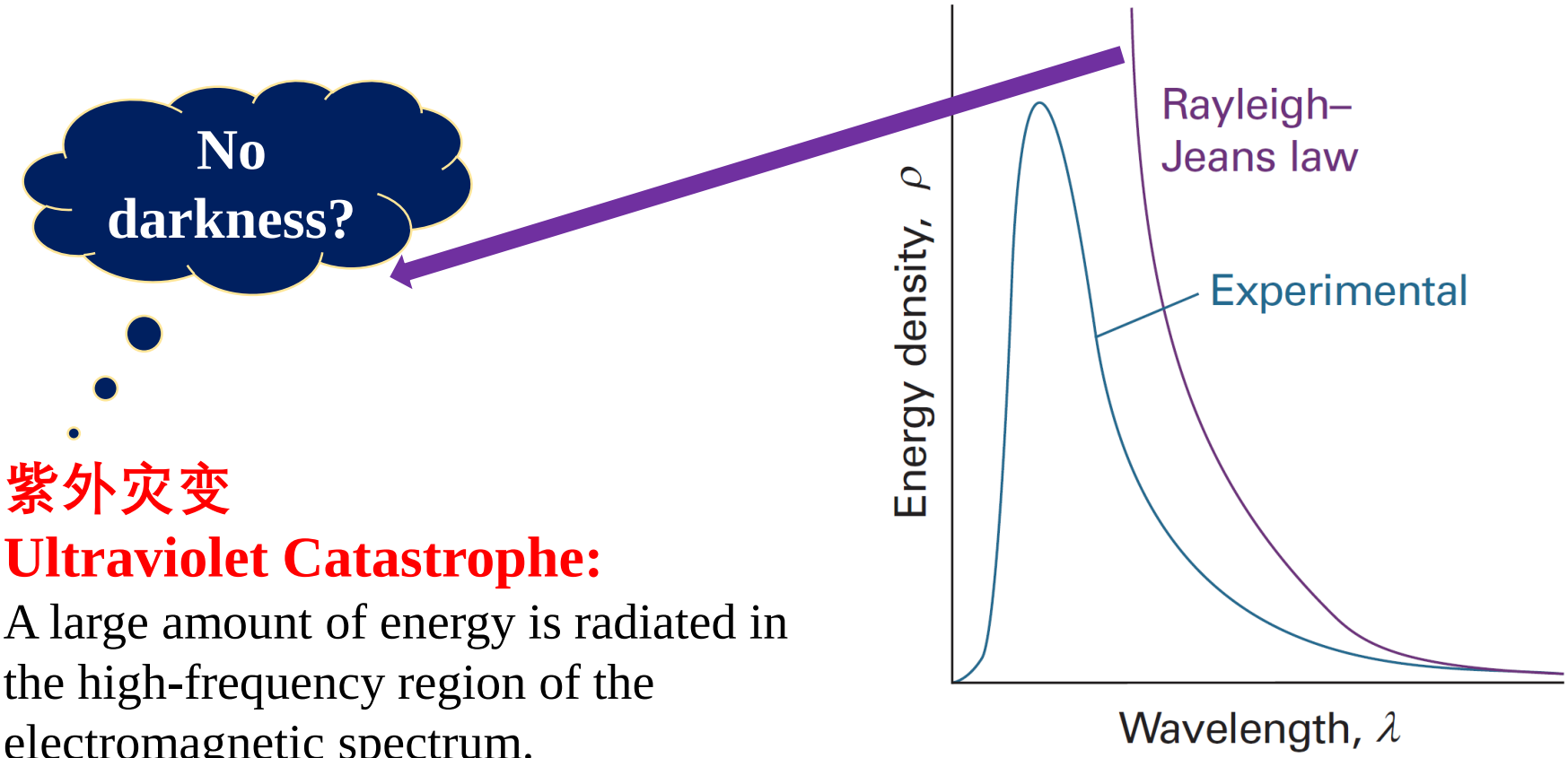
ElectroMagnetic Field : A collection of oscillators of all possible frequencies

Average energy of each oscillator: kT

Energy density $$\rho = \frac{8\pi kT}{\lambda^4}$$



Classical → Raleigh-Jeans law:

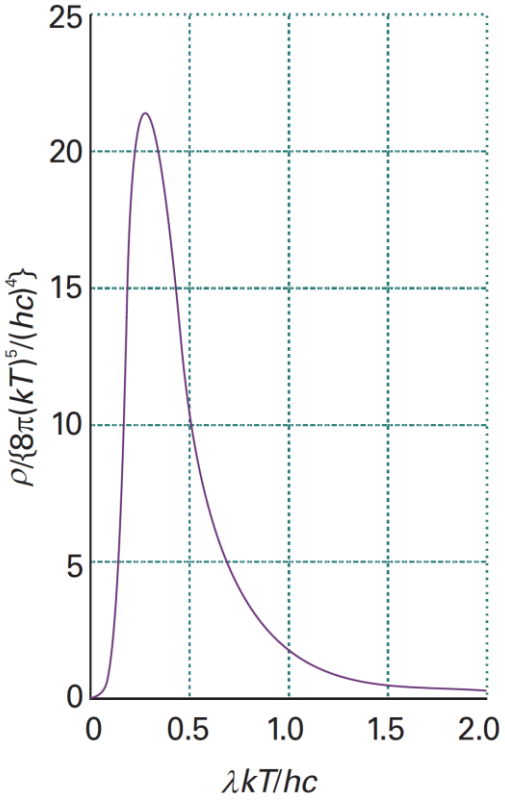


紫外灾变

Ultraviolet Catastrophe:

A large amount of energy is radiated in the high-frequency region of the electromagnetic spectrum.

Planck distribution → Quantum hypothesis



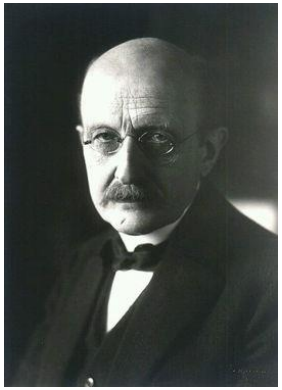
The energy of particles is **quantized**

$$E = nh\nu,\ n = 0, 1, 2, \ldots$$

ν is frequency

$h = 6.626\times10^{-34}$Js

Planck's constant



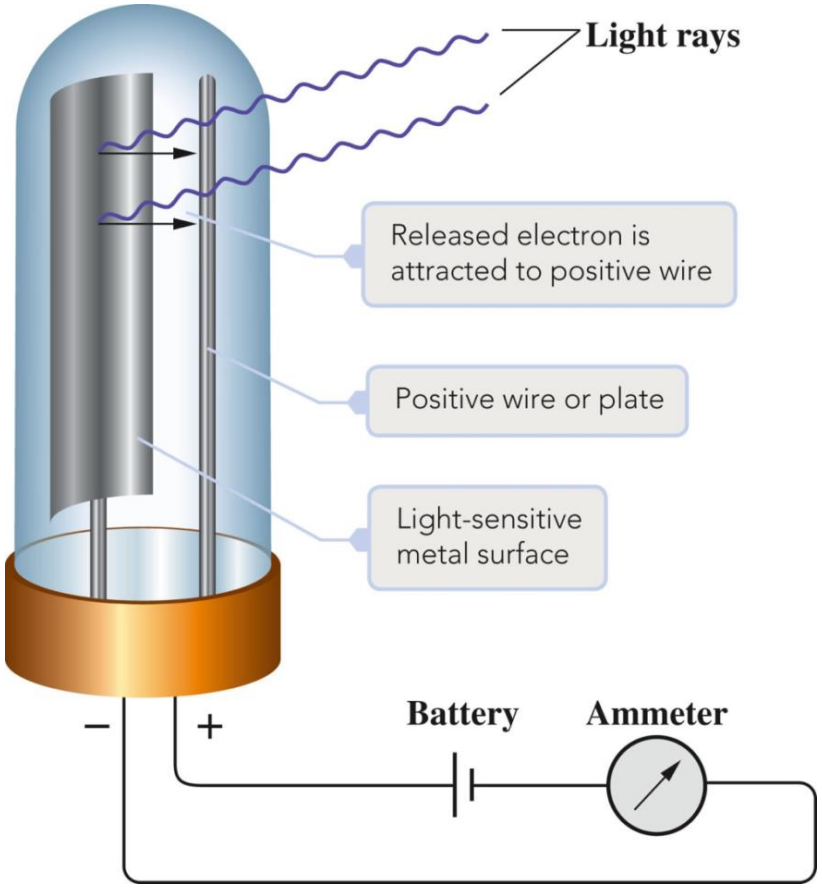
Max Planck
1858.4.23-1947.10.4

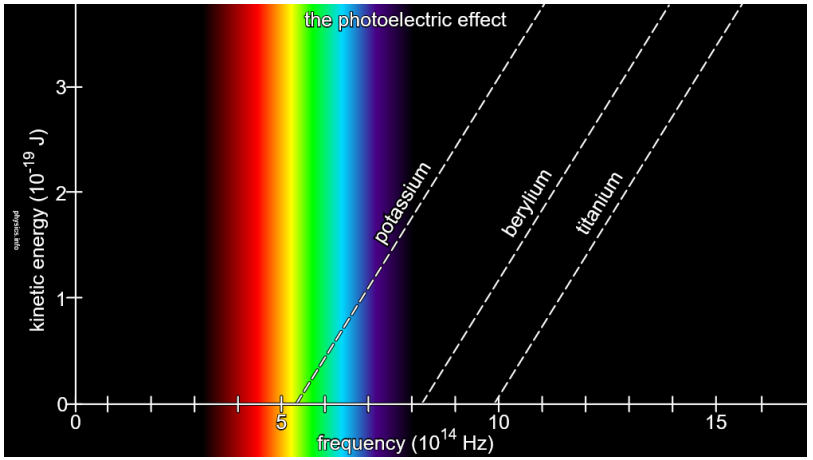
$$\rho = \frac{8\pi hc}{\lambda^5(e^{hc/\lambda kT} - 1)}$$

1918 Nobel Prize for Physics

Failures of classical physics

Photoelectric Effect: When ultraviolet light hits a metal surface, electrons are ejected only if light has a frequency above some threshold.





Kinetic energy of an emitted electron:

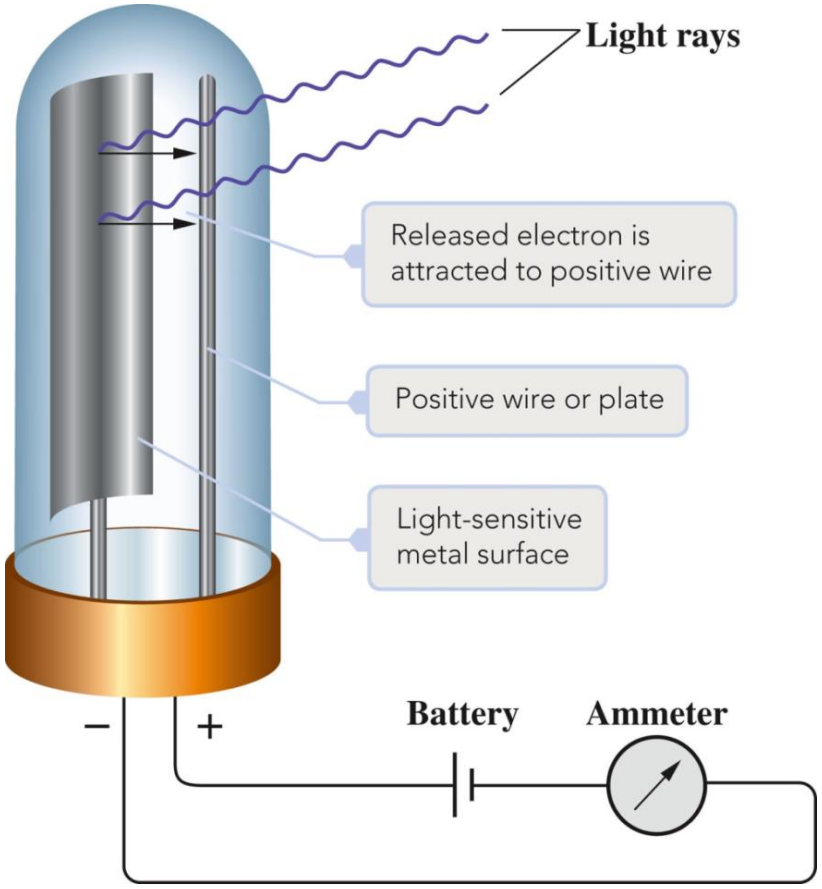
- Independent of the light's intensity
- Dependent of the light's frequency

For an electron to be emitted, the light must have $E > \Phi$

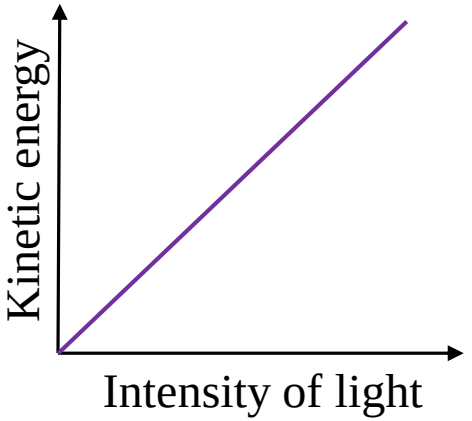
Work function

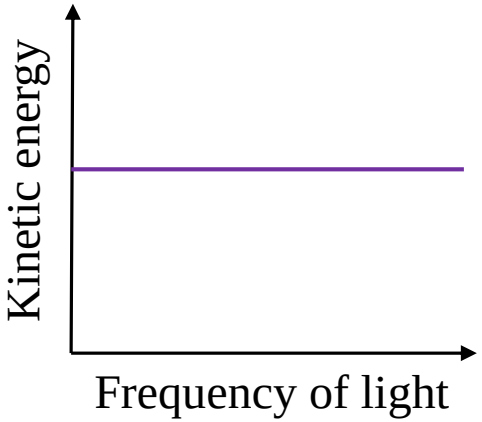
Failures of classical physics

Photoelectric Effect: When ultraviolet light hits a metal surface, electrons are ejected only if light has a frequency above some threshold.



Classical Mechanics





For an electron to be emitted, the light must have $E > \Phi$

Work function

Failures of classical physics

Photoelectric Effect: When ultraviolet light hits a metal surface, electrons are ejected only if light has a frequency above some threshold.

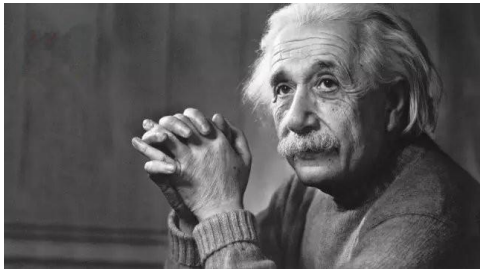
In 1905, Einstein proposed:

The electrons in the metal are interacting with individual particles of light (photons).

Light: composed of photons

Energy of photon: $\boldsymbol{E = h\nu}$

Albert Einstein
1879.3.14-1955.4.18



1921 Nobel Prize for Physics

Kinetic energy of an emitted electron:

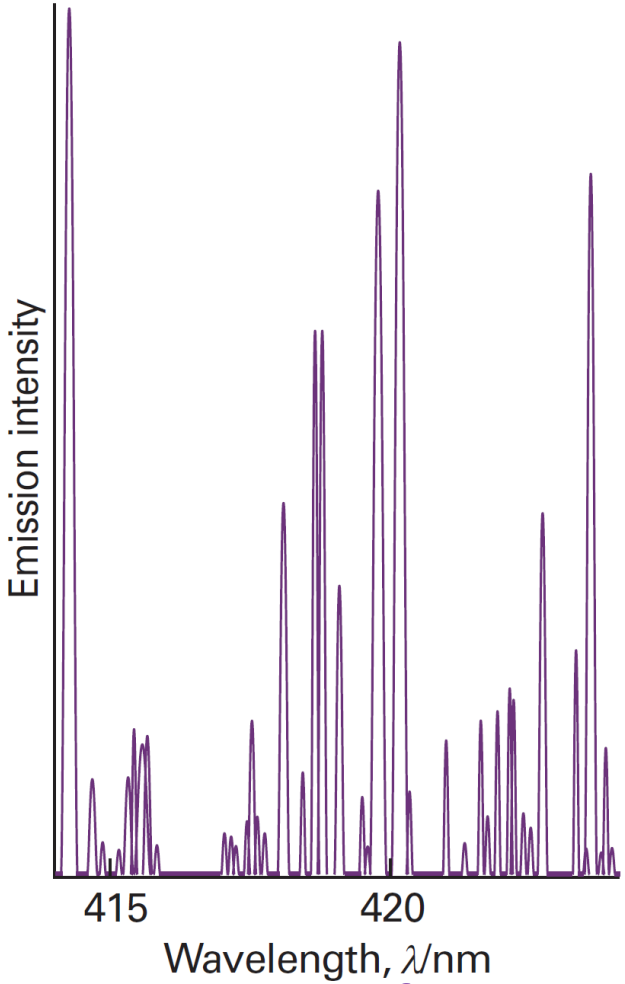
$$\frac{1}{2}m_e v^2 = h\nu - \Phi$$

$h = 6.626\times10^{-34}$Js

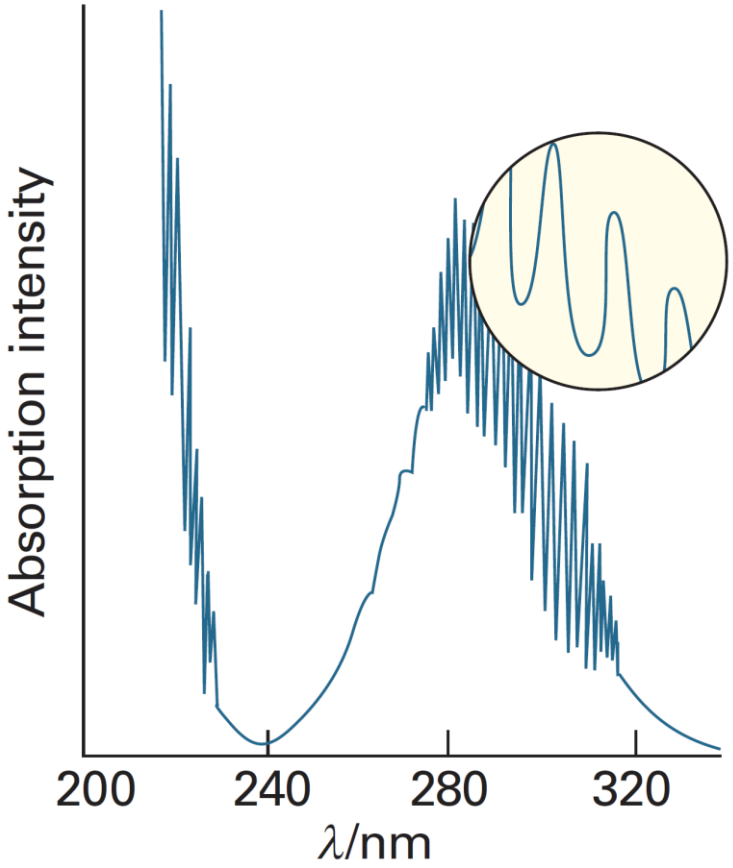
Work function
= Ionization energy

Atomic and molecular spectra

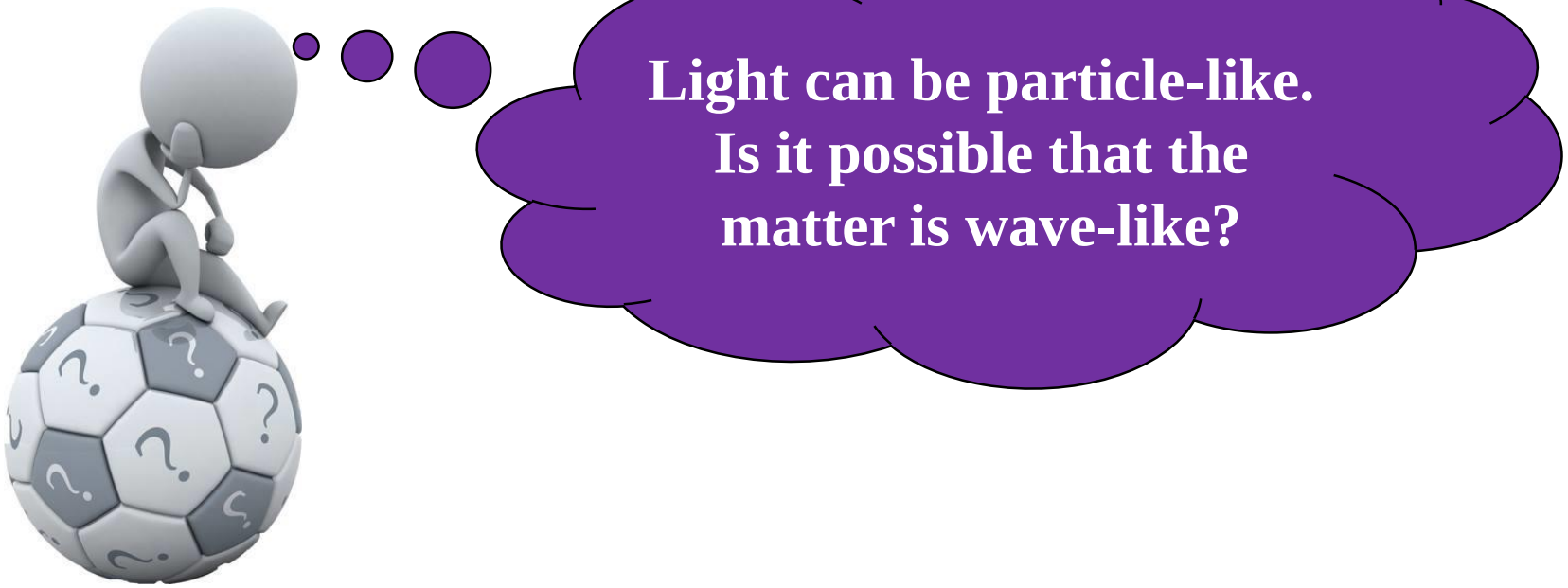
Radiation is emitted or absorbed at a series of discrete frequencies



Emission spectrum of excited iron atoms



Adsorption spectra of SO_2


Light can be particle-like.
Is it possible that the
matter is wave-like?

The wave character of particles

In 1924, Louis de Broglie (French):
Any particle, not only photons, travelling with a linear momentum p should have a wavelength given by **de Broglie relation**

$$\lambda = \frac{h}{p}$$

$h = 6.626\times10^{-34}$Js



Louis de Broglie
1892.8.15-1987.3.19

1929 Nobel Prize for Physics

This is by reasoning in analogy with photons.

The energy of a photon

$$E = mc^2 = h\nu = hc/\lambda \Rightarrow \lambda = h/mc = h/p$$

The wave character of particles

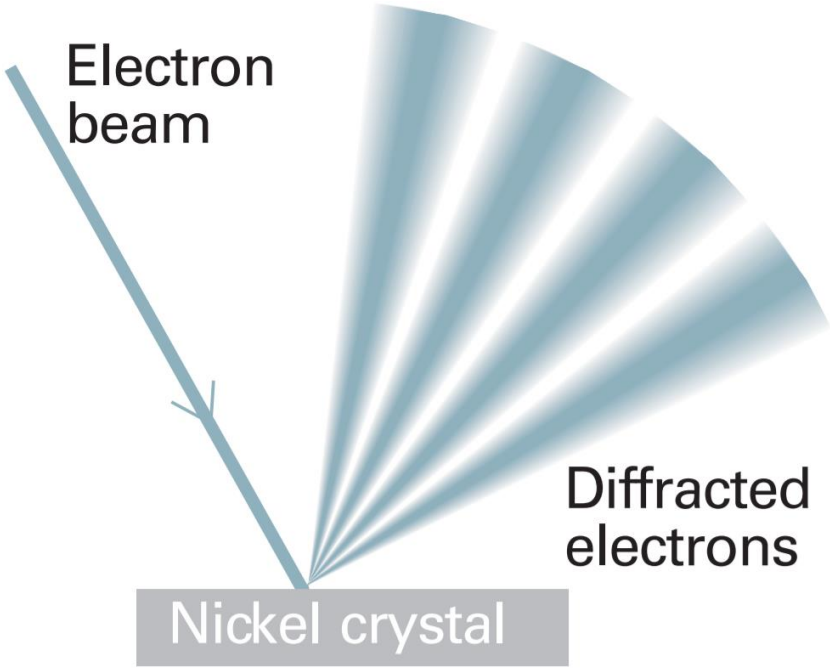


Clinton Joseph Davisson
1881.10.22-1958.2.01



George Paget Thomson
1892.5.3-1975.9.10

1937 Nobel Prize for Physics



Diffraction of electrons by crystals
(1927)

Example: Estimating the de Broglie wavelength

Estimate the wavelength of electrons that have been accelerated from rest through a potential difference of 40 kV.

Example: Estimating the de Broglie wavelength

Estimate the wavelength of electrons that have been accelerated from rest through a potential difference of 40 kV.

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad \lambda = \frac{h}{(2m_e eV)^{1/2}}$$

$$\lambda = \frac{6.626 \times 10^{-34}\ \text{J s}}{\{2 \times (9.109 \times 10^{-31}\ \text{kg}) \times (1.609 \times 10^{-19}\ \text{C}) \times (4.0 \times 10^{4}\ \text{V})\}^{1/2}}$$

$$= 6.1 \times 10^{-12}\ \text{m}$$

$$1\ \text{V C} = 1\ \text{J and } 1\ \text{J} = 1\ \text{kg m}^2\ \text{s}^{-2}$$

Wave-particle duality (波粒二象性) of matter

Heisenberg Uncertainty Principle:
It is impossible to specify simultaneously, with arbitrary precision, both the momentum and the position of a particle.

$$\Delta p \Delta q \geq \frac{1}{2}\hbar$$

p: the linear momentum
q: position
$\hbar = h/2\pi$



Werner Karl Heisenberg
1901.12.5-1976.2.1

1932 Nobel Prize for Physics
for the creation of quantum mechanics

Origin of Quantum Mechanics

Failures of classical physics to very small particles →

A new mechanics had to be devised for small objects!

→ **Quantum Mechanics**

$$H\Psi = E\Psi$$

Formulated by

- Heisenberg, Born, and Jordan in 1925 (***matrix mechanics***)
- Schrödinger in 1926 (***wave mechanics***)
- Dirac and Jordan in 1926 (***transformation theory***)
- Feynman in 1948 (***path integral formulation***)

Open Question Points: 10

Setting

Can you tell the difference between classical mechanics and quantum mechanics based on your understanding?

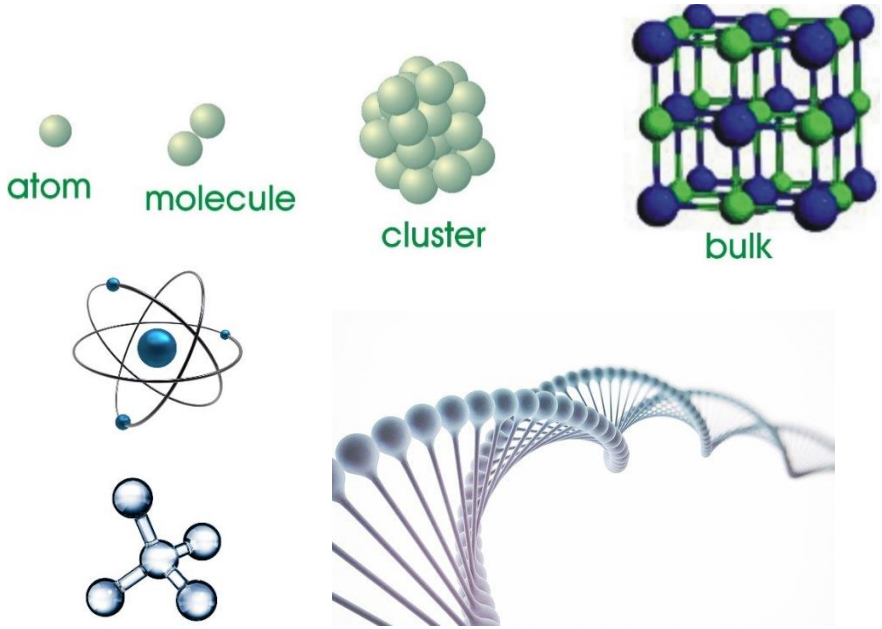
Open Question is only supported on Version 2.0 or newer.

Answer

Classical vs. Quantum

Classical Mechanics	Quantum Mechanics
Everyday-sized objects; **large, heavy**	Very, very **small** objects; **light**

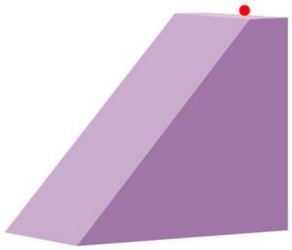




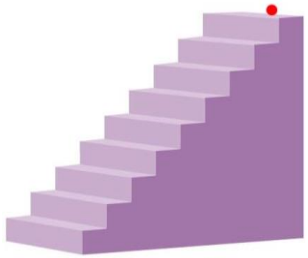
Classical vs. Quantum

Classical Mechanics	Quantum Mechanics
Continuous	Discrete/quantized

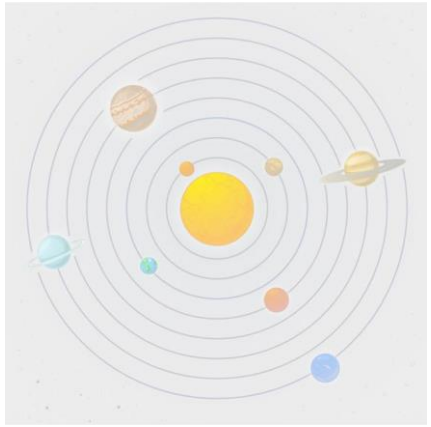


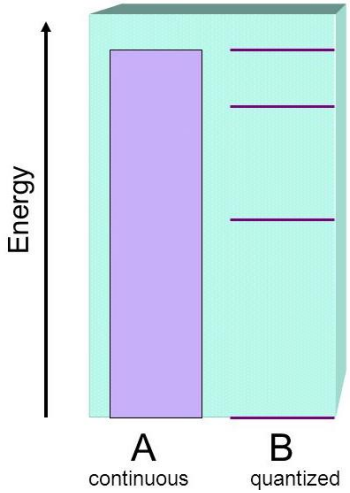


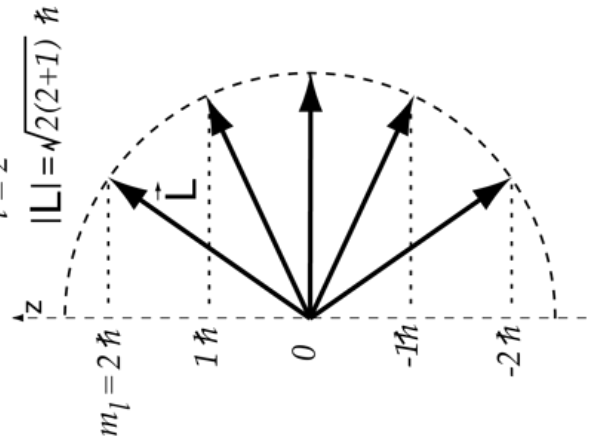






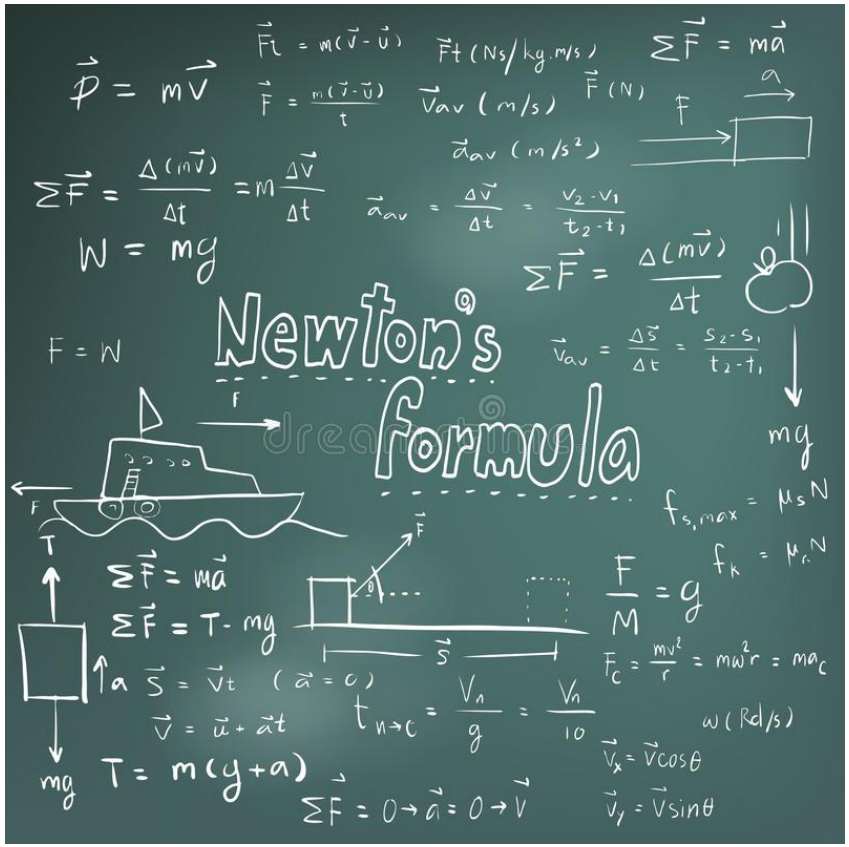


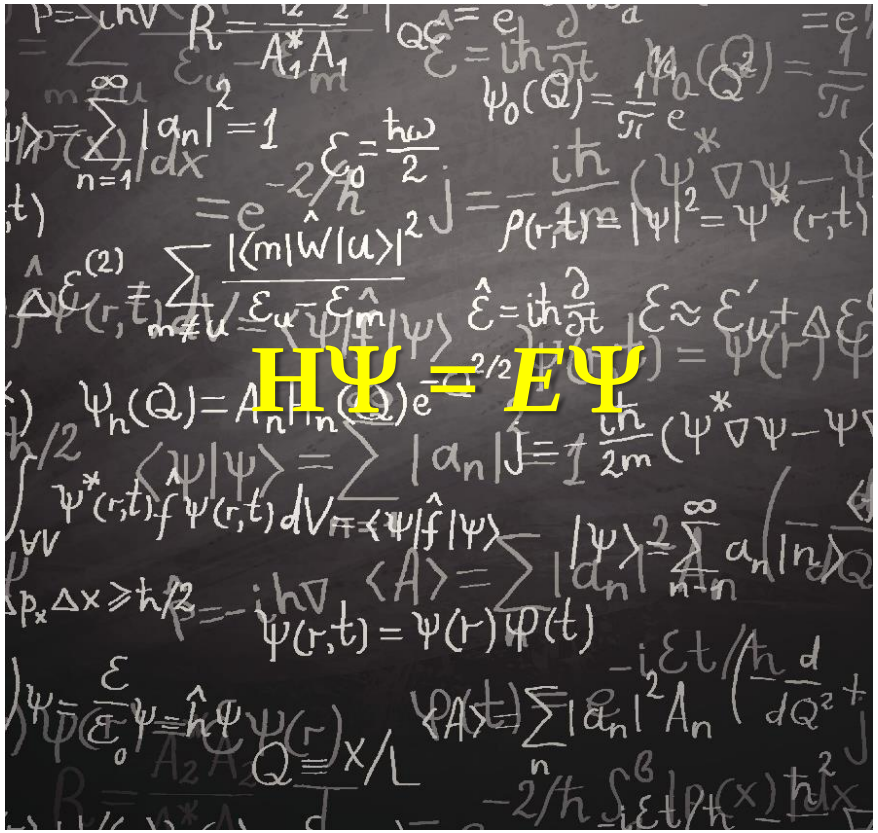




Classical vs. Quantum

Classical Mechanics	Quantum Mechanics
Newton's Eq.	**Schrödinger Eq.**

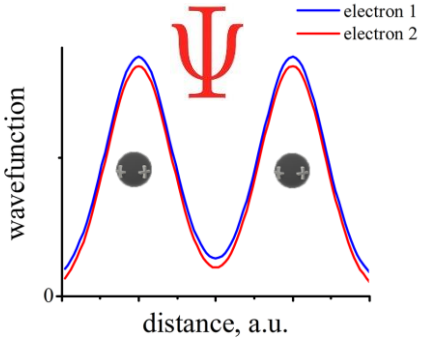


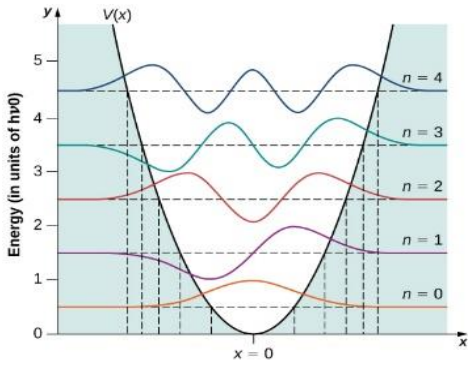


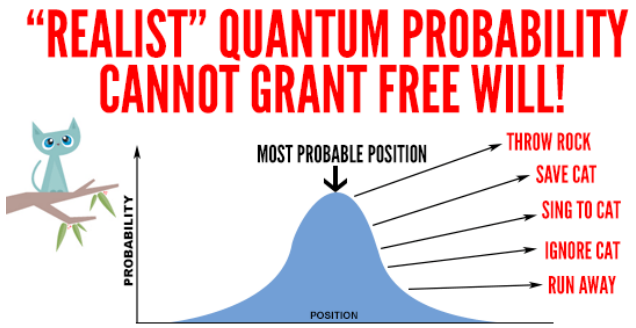
Classical vs. Quantum

Classical Mechanics	Quantum Mechanics
Trajectory； Deterministic	Wave function； Probabilistic











Classical vs. Quantum

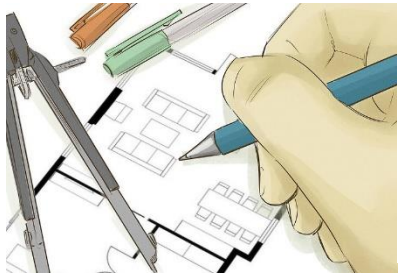
Classical Mechanics	Quantum Mechanics
Intuitive	**Non-intuitive**

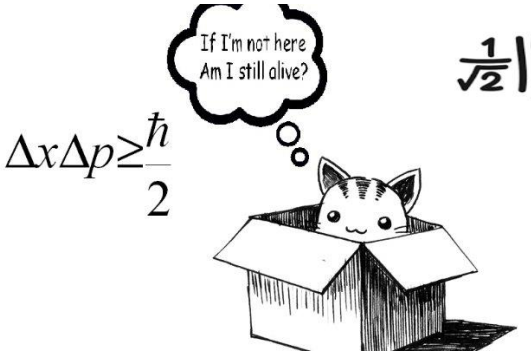




Counter intuitive:
Behave randomly
Somehow correlated two systems







Classical vs. Quantum

Classical Mechanics	Quantum Mechanics
Everyday-sized objects; large, heavy	Very, very small objects; light
Continuous	Discrete/quantized
Newton's Eq.	Schrödinger Eq.
Trajectory	Wave function
Deterministic	Probabilistic
Intuitive	Non-intuitive

Five minutes break!



The Schrödinger equation

Postulate 1. The state of a system is described by a function **Ψ** of the coordinates and the time. This function, called the state function or **wave function**, contains all the information that can be determined about the system.

Ψ(*q*,*t*): Single-valued, continuous, quadratically integrable

Postulate 6. The time development of the state of an undisturbed quantum-mechanical system is given by the **Schrödinger time-dependent equation**.

1933 Nobel Prize for Physics

$$-\frac{\hbar}{i}\frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

Hamiltonian operator



Erwin Schrödinger
1887.8.12-1961.1.4

For a one-particle, 1 D system:

$$-\frac{\hbar}{i}\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t)$$

$V(x,t)$: **Potential energy function**

$$i = \sqrt{-1} \qquad \hbar \equiv h/2\pi$$

$$\Psi(t_0) \rightarrow \Psi(t)$$

Ψ: $\rightarrow$ the definite specification of position? **NO!**

For a one-particle, 1 D system:

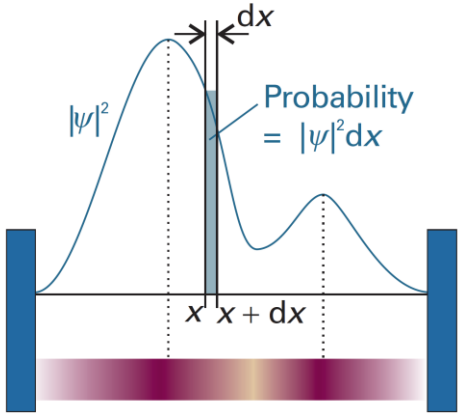
$$-\frac{\hbar}{i}\frac{\partial\Psi(x,t)}{\partial t}=-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2}+V(x,t)\Psi(x,t)$$



Max Born
1882.12.11-
1970.1.5

Born interpretation:

$|\Psi(x,t)|^2dx$ gives the **probability** at time t of finding the particle in the region of the x axis lying between x and $x + dx$.



1954 Nobel Prize for Physics

Probability density

for finding the particle at various places on the x axis at time t.

$$\int_{-\infty}^{\infty}|\psi(x)|^2dx=1$$

Normalized！归一的！

For a one-particle, 1 D system:

Schrödinger time-dependent equation

$$-\frac{\hbar}{i}\frac{\partial\Psi(x,t)}{\partial t}=-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2}+V(x,t)\Psi(x,t)$$



Formidable?

Fortunately, many applications of quantum mechanics to chemistry use the simpler **time-independent Schrodinger equation**!



A special 1-D case, $V(x,t) = V(x)$

(A system experiencing no time-dependent external forces)

Look for the solutions that can be written as

$$\Psi(x,t) = f(t)\psi(x)$$

!!!Not all solutions have this form!

$$\frac{\partial \Psi(x,t)}{\partial t} = \frac{df(t)}{dt}\psi(x)$$

$$\frac{\partial^2 \Psi(x,t)}{\partial x^2} = f(t)\frac{d^2\psi(x)}{dx^2}$$

A special 1-D case, $V(x,t) = V(x)$

$$-\frac{\hbar}{i}\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t)$$



$$-\frac{\hbar}{i}\frac{df(t)}{dt}\psi(x) = -\frac{\hbar^2}{2m} f(t)\frac{d^2\psi(x)}{dx^2} + V(x) f(t)\psi(x)$$



Divided by $f\psi$

$$-\frac{\hbar}{i}\frac{1}{f(t)}\frac{df(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + V(x) \quad = \boldsymbol{E}$$

***E* is a constant**

A special 1-D case, $V(x,t) = V(x)$

$$-\frac{\hbar}{i}\frac{1}{f(t)}\frac{df(t)}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi(x)}\frac{d^2\psi(x)}{dx^2} + V(x)$$

Equating the left side to E

$$\frac{df(t)}{f(t)} = -\frac{iE}{\hbar}dt$$

a constant

- $\ln f(t) = -\frac{iEt}{\hbar} + C$
- $f(t) = e^{C}e^{-iEt/\hbar} = Ae^{-\frac{iEt}{\hbar}}$
- $f(t) = e^{-iEt/\hbar}$
- $\Psi(x,t) = e^{-iEt/\hbar}\psi(x)$

Equating the right side to E

The time-independent Schrödinger equation :

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

Postulate: ***E* is the energy of the system!**

A special 1-D case, $V(x,t) = V(x)$

The time-independent Schrödinger equation :

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}+V(x)\psi(x)=E\psi(x)$$

$$\Psi(x,t)=e^{-iEt/\hbar}\psi(x)$$

$$\longrightarrow \quad \left|\Psi(x,t)\right|^2=\Psi^*(x,t)\Psi(x,t)=\left|\psi(x)\right|^2$$

Stationary state (定态):

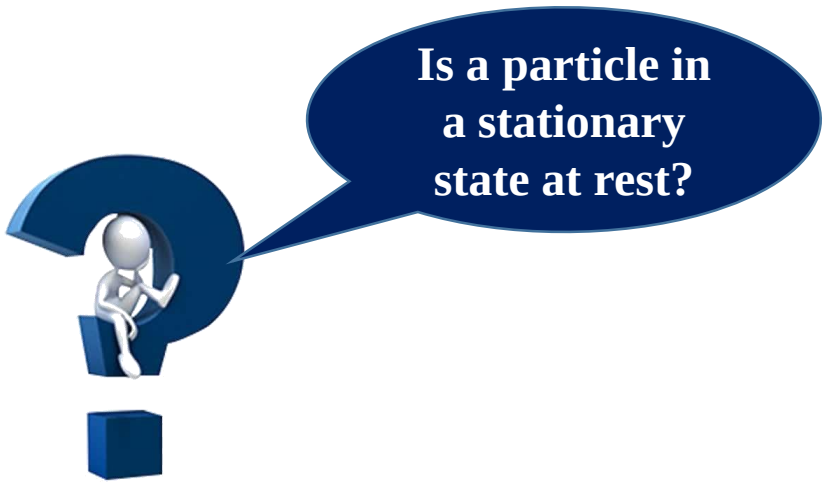
Probability density does not change with time.

A special 1-D case, $V(x,t) = V(x)$

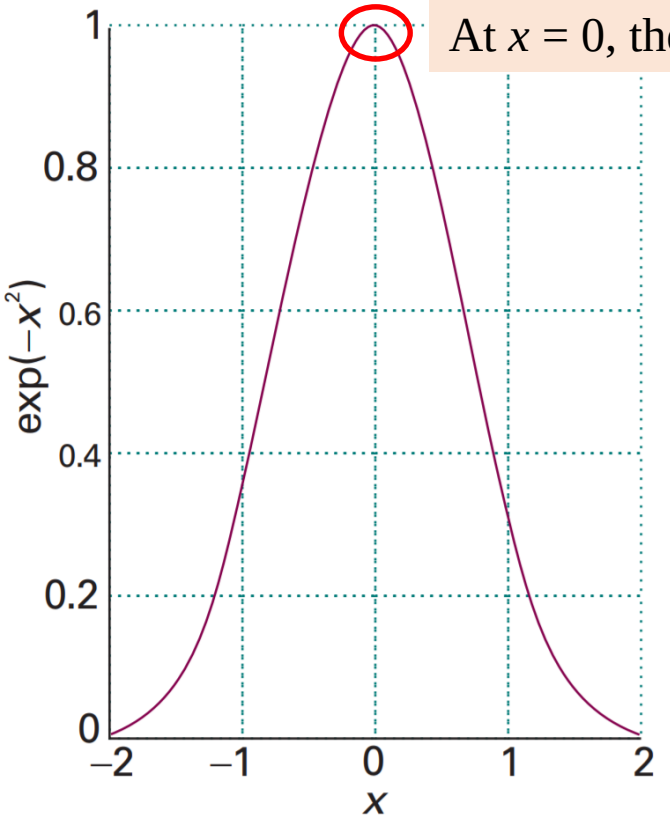
The time-independent Schrödinger equation :

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}+V(x)\psi(x)=E\psi(x)$$

$$\Psi(x,t)=e^{-iEt/\hbar}\psi(x)$$







At $x = 0$, the particle is more likely to be found

A particle at a state $\psi = ae^{-bx^2}$

Probability density $|\psi|^2 = a^2 e^{-2bx^2}$

Quantum mechanics is basically statistical in nature!

Two minutes break!



Operators 算符，算子

Schrödinger equation : (1-D)

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}+V(x)\psi(x)=E\psi(x)$$

$$\hat{H}\psi=E\psi$$

Hamiltonian Operator
哈密顿算子

$$\hat{H}=-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)$$

One **operator** is something that carries out a mathematical operation on a function; a rule that transforms a given function into another function.

Some operators:

$$\hat{A} = \frac{d}{dx}, \qquad \hat{A}f = \frac{df}{dx}$$

$$\hat{B} = x, \qquad \hat{B}f = xf$$

$$\hat{c} = c, \qquad \hat{c}f = cf$$

$$\hat{1} = 1$$ **Unit operator**

$$\hat{0} = 0$$ **Null operator**

Hamiltonian Operator

In 1-D system



William Rowan Hamilton
1805.8.4-1865.9.2

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

Total energy E

Kinetic energy $\hat{T}$

Potential energy $\hat{V}(x)$

Kinetic energy operator $\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$

Kinetic energy $E_k = \frac{1}{2}mv_x^2 = \frac{p_x^2}{2m}$

Correspondence

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x} = -i\hbar\frac{\partial}{\partial x}$$

Linear Momentum Operator

$$\hat{x} = x$$

Operator of Cartesian coordinate *x*

Properties in C.M. → **Operators in Q.C.**

Postulate 2. To every physically observable property there corresponds a **linear Hermitian operator**. To find this operator, write down the classical-mechanical expression for the observable in terms of Cartesian coordinates and corresponding linear-momentum components, and then replace each coordinate x by the operator $x\bullet$ and each momentum component p_x by the operator $-i\hbar\partial/\partial x$.

Physical quantity	Operator
Position *x*	$\hat{x} = x$
The *x* component of linear momentum p_x	$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$
Kinetic energy *T*	$\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$
Potential energy *V*	$\hat{V}(x) = V(x)$
Total energy *E*	$\hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$

Operator Algebra

Sum and difference of two operators

$$(\hat{A}+\hat{B})\,f(x) \equiv \hat{A}f(x)+\hat{B}f(x)$$

$$(\hat{A}-\hat{B})\,f(x) \equiv \hat{A}f(x)-\hat{B}f(x)$$

Product of two operators

$$(\hat{A}\hat{B})\,f(x) \equiv \hat{A}\left[\,\hat{B}f(x)\right]$$

Square of an operator

$$\hat{A}^2 = \hat{A}\hat{A}$$

Obey the associative law of multiplication

$$\hat{A}\left(\hat{B}\hat{C}\right) = \left(\hat{A}\hat{B}\right)\hat{C}$$

Open Question Points: 10

Setting

$$\hat{A}\hat{B} \overset{?}{=} \hat{B}\hat{A}$$

(1) $\hat{A} = \dfrac{d}{dx}, \quad \hat{B} = x,$

(2) $\hat{A} = \dfrac{d}{dx}, \quad \hat{B} = 3,$

Open Question is only supported on Version 2.0 or newer.

Answer

$$\hat{A}\hat{B} \stackrel{?}{=} \hat{B}\hat{A}$$

Commutator 对易子

$$\left[\hat{A},\hat{B}\right] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$$

Commute 对易

$$\left[\hat{A},\hat{B}\right] = 0$$

Calculate commutator $\left[\hat{A}, \hat{B}\right] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$

(1) $\hat{A} = \frac{d}{dx}, \quad \hat{B} = x,$

(2) $\hat{A} = \frac{d}{dx}, \quad \hat{B} = 3,$

Linear (线性) Operator:

$$\hat{A}[f(x)+g(x)]=\hat{A}f(x)+\hat{A}g(x)$$

$$\hat{A}[cf(x)]=c\hat{A}f(x)$$

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Which are linear operators? Which are not? Why?

$$\hat{A}=\frac{d}{dx},\ \hat{B}=x^2,\hat{C}=\frac{d^2}{dx^2},$$
$$\hat{D}=\cos,\hat{E}=(\ )^2$$

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Useful identities (恒等式) in linear operator

$$\left(\hat{A}+\hat{B}\right)\hat{C}=\hat{A}\hat{C}+\hat{B}\hat{C}$$

$$\hat{A}\left(\hat{B}+\hat{C}\right)=\hat{A}\hat{B}+\hat{A}\hat{C}$$

Hermitian (厄米) Operator is a linear operator that satisfies

$$\int \psi_i^* \hat{\Omega} \psi_j dx = \left\{ \int \psi_j^* \hat{\Omega} \psi_i dx \right\}^*$$

Or $$\int f^* \hat{\Omega} g d\tau = \int g \left(\hat{\Omega} f \right)^* d\tau$$ for all well-behaved functions

The position operator x is hermitian

$$\int_{-\infty}^{\infty} \psi_i^* x \psi_j dx = \int_{-\infty}^{\infty} \psi_j x \psi_i^* dx = \left\{ \int_{-\infty}^{\infty} \psi_j^* x \psi_i dx \right\}^*$$

Justification: *The hermiticity of the linear momentum operator*

"integration by parts"

$$\int f\frac{\mathrm{d}g}{\mathrm{d}x}\mathrm{d}x = fg - \int g\frac{\mathrm{d}f}{\mathrm{d}x}\mathrm{d}x$$

Justification: *The hermiticity of the linear momentum operator*

Our task is to show that

$$\int_{-\infty}^{\infty} \psi_i^{\star} \hat{p}_x \psi_j \mathrm{d}x = \left\{ \int_{-\infty}^{\infty} \psi_j^{\star} \hat{p}_x \psi_i \mathrm{d}x \right\}^{\star}$$

"integration by parts"

$$\int f \frac{\mathrm{d}g}{\mathrm{d}x} \mathrm{d}x = fg - \int g \frac{\mathrm{d}f}{\mathrm{d}x} \mathrm{d}x$$

$$\int_{-\infty}^{\infty} \psi_i^{\star} \hat{p}_x \psi_j \mathrm{d}x = \frac{\hbar}{\mathrm{i}} \int_{-\infty}^{\infty} \psi_i^{\star} \frac{\mathrm{d}\psi_j}{\mathrm{d}x} \mathrm{d}x$$

$$= \underbrace{\frac{\hbar}{\mathrm{i}} \psi_i^{\star} \psi_j \Big|_{-\infty}^{\infty}} - \frac{\hbar}{\mathrm{i}} \int_{-\infty}^{\infty} \psi_j \frac{\mathrm{d}\psi_i^{\star}}{\mathrm{d}x} \mathrm{d}x$$

= 0, because well-behaved wavefunctions are 0 at infinity in either direction

$$\int_{-\infty}^{\infty} \psi_i^{\star} \hat{p}_x \psi_j \mathrm{d}x = -\frac{\hbar}{\mathrm{i}} \int_{-\infty}^{\infty} \psi_j \frac{\mathrm{d}\psi_i^{\star}}{\mathrm{d}x} \mathrm{d}x = \left\{ \frac{\hbar}{\mathrm{i}} \int_{-\infty}^{\infty} \psi_j^{\star} \frac{\mathrm{d}\psi_i}{\mathrm{d}x} \mathrm{d}x \right\}^{\star}$$

$$= \left\{ \int_{-\infty}^{\infty} \psi_j^{\star} \hat{p}_x \psi_i \mathrm{d}x \right\}^{\star}$$